

Unit 4

Network Layer

- The Network Layer is the third layer of the OSI model.
- It handles the service requests from the transport layer and further forwards the service request to the data link layer.
- The network layer translates the logical addresses into physical addresses
- It determines the route from the source to the destination and also manages the traffic problems such as switching, routing and controls the congestion of data packets.
- The main role of the network layer is to move the packets from sending host to the receiving host.

The main functions performed by the network layer are:

- **Routing:** When a packet reaches the router's input link, the router will move the packets to the router's output link. For example, a packet from S1 to R1 must be forwarded to the next router on the path to S2.
- **Logical Addressing:** The data link layer implements the physical addressing and network layer implements the logical addressing. Logical addressing is also used to distinguish between source and destination system. The network layer adds a header to the packet which includes the logical addresses of both the sender and the receiver.
- **Internetworking:** This is the main role of the network layer that it provides the logical connection between different types of networks.
- **Fragmentation:** The fragmentation is a process of breaking the packets into the smallest individual data units that travel through different networks.

- **IPV4**

IP stands for Internet Protocol and v4 stands for Version Four (IPv4). IPv4 was the primary version brought into action for production within the ARPANET in 1983.

IP version four addresses are 32-bit integers which will be expressed in decimal notation.

Example- 192.0.2.126 could be an IPv4 address.

Parts of IPv4

Network part:

The network part indicates the distinctive variety that's appointed to the network. The network part conjointly identifies the category of the network that's assigned.

Host Part:

The host part uniquely identifies the machine on your network. This part of the IPv4 address is assigned to every host. For each host on the network, the network part is the same, however, the host half must vary.

Subnet number:

This is the nonobligatory part of IPv4. Local networks that have massive numbers of hosts are divided into subnets and subnet numbers are appointed to that.

Characteristics of IPv4

1. IPv4 could be a 32-Bit IP Address.
2. IPv4 could be a numeric address, and its bits are separated by a dot.
3. The number of header fields is twelve and the length of the header field is twenty.
4. It has Unicast, broadcast, and multicast style of addresses.
5. IPv4 supports VLSM (Virtual Length Subnet Mask).
6. IPv4 uses the Post Address Resolution Protocol to map to the MAC address.
7. RIP may be a routing protocol supported by the routed daemon.
8. Networks ought to be designed either manually or with DHCP.
9. Packet fragmentation permits from routers and causing host.

Advantages of IPv4

1. IPv4 security permits encryption to keep up privacy and security.
2. IPV4 network allocation is significant and presently has quite 85000 practical routers.
3. It becomes easy to attach multiple devices across an outsized network while not NAT.
4. This is a model of communication so provides quality service also as economical knowledge transfer.
5. IPV4 addresses are redefined and permit flawless encoding.
6. Routing is a lot of scalable and economical as a result of addressing is collective more effectively.
7. Data communication across the network becomes a lot of specific in multicast organizations.
8. Limits net growth for existing users and hinders the use of the net for brand new users.
9. Internet Routing is inefficient in IPv4.
10. IPv4 has high System Management prices and it's labor-intensive, complex, slow & frequent to errors.
11. Security features are nonobligatory.

• **IPV6**

IPv6 (Internet Protocol version 6) is the most recent version of the Internet Protocol, designed as the successor to IPv4. It was developed to address the limitations of IPv4, particularly in terms of addressing, scalability, and security. Here are some key points about IPv6:

1. Addressing: IPv6 uses a 128-bit address space, allowing for a significantly larger number of unique IP addresses compared to IPv4. IPv6 addresses are written in hexadecimal notation and are represented as eight groups of four hexadecimal digits, separated by colons. For example, 2001:0db8:85a3:0000:0000:8a2e:0370:7334.
2. Address Format: IPv6 addresses are hierarchical and typically consist of a network prefix and an interface identifier. The network prefix identifies the network portion of the address, while the interface identifier identifies the specific device on the network. IPv6 also introduces special address types, such as link-local addresses and multicast addresses.
3. Address Autoconfiguration: IPv6 includes built-in support for address autoconfiguration through the Stateless Address Autoconfiguration (SLAAC) mechanism. With SLAAC, devices on a network can generate their own IPv6 addresses using the network prefix provided by a router. This simplifies network administration and eliminates the need for DHCP (Dynamic Host Configuration Protocol) in many cases.

4. **Larger Address Space:** IPv6's larger address space allows for the allocation of unique addresses to an ever-growing number of devices, including computers, smartphones, Internet of Things (IoT) devices, and more. This helps address the address exhaustion issues faced by IPv4.
5. **Enhanced Security:** IPv6 incorporates IPsec (Internet Protocol Security) as a mandatory part of the protocol suite. IPsec provides encryption, authentication, and integrity verification of IP packets, improving security at the network layer.
6. **Transition Mechanisms:** IPv6 adoption has been gradual, and during the transition period, mechanisms such as dual-stack operation, tunneling (e.g., 6to4, Teredo), and translation (e.g., NAT64) have been developed to facilitate communication between IPv4 and IPv6 networks.
7. **Deployment and Support:** IPv6 is supported by most modern operating systems, network devices, and Internet service providers (ISPs). Major websites and online services are increasingly adopting IPv6 to ensure compatibility and meet the growing demand for IP addresses.
8. **Benefits:** The adoption of IPv6 brings several benefits, including a virtually unlimited address space, improved network performance, enhanced security, simplified network administration, and support for new services and devices in the evolving Internet landscape.

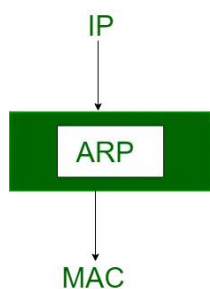
While IPv4 is still widely used, the transition to IPv6 is gradually taking place to accommodate the increasing demands of the Internet and the growing number of connected devices. IPv6 provides the necessary infrastructure to support the future growth of the Internet and ensure its continued functioning.

- **Address Mapping**

Address mapping is a process of determining a logical address knowing the physical address of the device and determining the physical address by knowing the logical address of the device. Address mapping is required when a packet is routed from source host to destination host in the same or different network.

- **ARP**

Most computer programs/applications use [logical addresses \(IP Addresses\)](#) to send/receive messages. However, the actual communication happens over the Physical Address (MAC Address) that is from layer 2 of the OSI model. So our mission is to get the destination MAC Address which helps in communicating with other devices. This is where ARP comes into the picture, its functionality is to translate IP addresses to Physical Addresses.

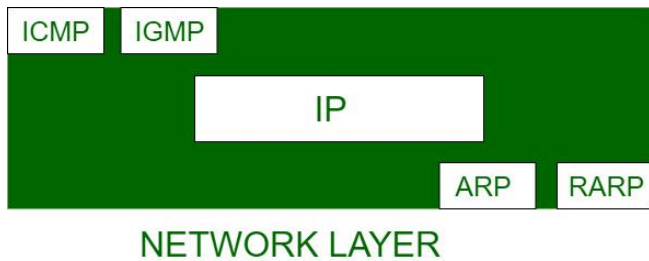


ARP

Address Resolution Protocol (ARP)

The acronym ARP stands for Address Resolution Protocol which is one of the most important protocols of the Network layer in the OSI model. It is responsible to find the hardware address of a host from a known IP address. There are three basic ARP terms.

Note: ARP finds the hardware address, also known as the [Media Access Control \(MAC\)](#) address, of a host from its known IP address.



Address Resolution Protocol

Important Terms Associated with ARP

[Reverse ARP](#)

[Proxy ARP](#)

[Inverse ARP](#)

Reverse ARP

Reverse Address Resolution Protocol is a protocol that is used in local area networks (LAN) by client machines for requesting IP Address (IPv4) from Router's ARP Table. Whenever a new machine comes, which requires an IP Address for its use. In that case, the machine sends a RARP broadcast packet containing MAC Address in the sender and receiver hardware field.

Proxy ARP

Proxy Address Resolution Protocol work to enable devices that are separated into network segments connected through the router in the same IP to resolve IP Address to MAC Address. Proxy ARP is enabled so that the 'proxy router' resides with its MAC address in a local network as it is the desired router to which broadcast is addressed. In case, when the sender receives the MAC Address of the Proxy Router, it is going to send the datagram to Proxy Router, which will be sent to the destination device.

Inverse ARP

Inverse Address Resolution Protocol uses MAC Address to find the IP Address, it can be simply illustrated as Inverse ARP is just the inverse of ARP. In ATM (Asynchronous Transfer Mode) Networks, Inverse ARP is used by default. Inverse ARP helps in finding Layer-3 Addresses from Layer-2 Addresses.

- **RARP**

The Reverse Address Resolution Protocol (RARP) is a networking protocol that is used to map a physical (MAC) address to an Internet Protocol (IP) address. It is the reverse of the more commonly used Address Resolution Protocol (ARP), which maps an IP address to a MAC address.

RARP was developed in the early days of computer networking as a way to provide IP addresses to diskless workstations or other devices that could not store their own IP addresses. With RARP, the device would broadcast its MAC address and request an IP address, and a RARP server on the network would respond with the corresponding IP address.

While RARP was widely used in the past, it has largely been replaced by newer protocols such as DHCP (Dynamic Host Configuration Protocol), which provides more flexibility and functionality in assigning IP addresses dynamically. However, RARP is still used in some specialized applications, such as booting embedded systems and configuring network devices with pre-assigned IP addresses.

- **DHCP**

DHCP stands for Dynamic Host Configuration Protocol. It is the critical feature on which the users of an enterprise network communicate. DHCP helps enterprises to smoothly manage the allocation of [IP addresses](#) to the end-user clients' devices such as desktops, laptops, cellphones, etc. is an application layer protocol that is used to provide:

Subnet Mask (Option 1 - e.g., 255.255.255.0)

Router Address (Option 3 - e.g., 192.168.1.1)

DNS Address (Option 6 - e.g., 8.8.8.8)

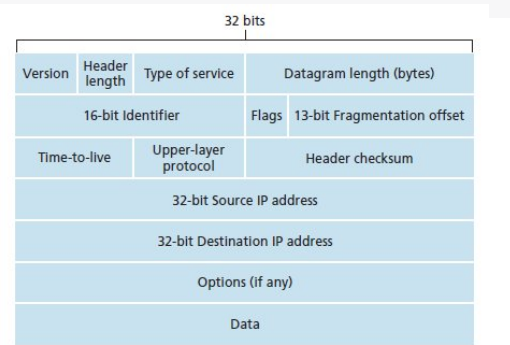
Vendor Class Identifier (Option 43 - e.g.,
'unifi' = 192.168.1.9 ##where unifi = controller)

DHCP is based on a [client-server model](#) and based on discovery, offer, request, and ACK.

DHCP helps in managing the entire process automatically and centrally. DHCP helps in maintaining a unique IP Address for a host using the server. DHCP servers maintain information on TCP/IP configuration and provide configuration of address to DHCP-enabled clients in the form of a lease offer.

- **IPv4 datagram detail format**

The IPv4 datagram is the packet format used in the IPv4 protocol for transmitting data across networks. It consists of various fields that carry information about the source and destination IP addresses, protocol type, packet length, fragmentation, and other control information. Here's a breakdown of the IPv4 datagram format:



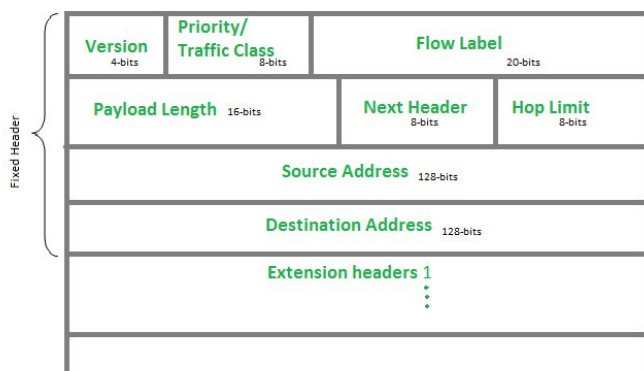
1. Version (4 bits):
 2. IHL (Internet Header Length) (4 bits):
 3. Type of Service (8 bits):
- The Version field indicates the IP version being used, which is IPv4 in this case. It is set to 4 for IPv4.

The IHL field specifies the length of the IP header in 32-bit words. Since the IPv4 header has a variable length, this field helps identify the start of the data portion. The value in the IHL field is the number of 32-bit words, so a value of 5 indicates a 20-byte header (5 * 4 = 20 bytes).

The Type of Service field carries information related to Quality of Service (QoS) and prioritization of the IP packet. It includes subfields like Differentiated Services Code Point (DSCP) and Explicit Congestion Notification (ECN).

4. Total Length (16 bits): The Total Length field specifies the total length of the IP datagram (header + data) in bytes. It indicates the maximum size of the datagram that can be transmitted in one piece. The maximum value is 65,535 bytes.
5. Identification (16 bits): The Identification field is used to uniquely identify a fragmented IP datagram. It remains the same across all fragments of a particular IP datagram.
6. Flags (3 bits): The Flags field contains control bits related to IP fragmentation. The three flags are:
 - Reserved (always set to 0)
 - Don't Fragment (DF): If set to 1, indicates that the packet should not be fragmented.
 - More Fragments (MF): If set to 1, indicates that there are more fragments following the current fragment.
7. Fragment Offset (13 bits): The Fragment Offset field indicates the position of the current fragment within the original IP datagram, measured in units of 8 bytes.
8. Time to Live (TTL) (8 bits): The Time to Live field specifies the maximum number of network hops the datagram can take before being discarded. Each router decreases this value by 1, and if it reaches 0, the datagram is discarded.
9. Protocol (8 bits): The Protocol field identifies the upper-layer protocol (e.g., TCP, UDP) to which the IP packet is handed off after processing the IP layer.
10. Header Checksum (16 bits): The Header Checksum field is used to ensure the integrity of the IP header during transmission. It is computed over the entire header and checked by the recipient.
11. Source IP Address (32 bits): The Source IP Address field contains the IP address of the sender (source) of the IP datagram.
12. Destination IP Address (32 bits): The Destination IP Address field contains the IP address of the intended recipient (destination)

- **IP version 6 Header Format:**



Version (4-bits): Indicates version of Internet Protocol which contains bit sequence 0110.

Traffic Class (8-bits): The Traffic Class field indicates class or priority of IPv6 packet which is similar to Service Field in IPv4 packet. It helps routers to handle the traffic based on the priority of the packet. If congestion occurs on the router then packets with the least priority will be discarded.

As of now, only 4-bits are being used (and the remaining bits are under research), in which 0 to 7 are assigned to Congestion controlled traffic and 8 to 15 are assigned to Uncontrolled traffic.

Priority assignment of Congestion controlled traffic :

Priority	Meaning
0	No Specific traffic
1	Background data
2	Unattended data traffic
3	Reserved
4	Attended bulk data traffic
5	Reserved
6	Interactive traffic
7	Control traffic

Uncontrolled data traffic is mainly used for Audio/Video data. So we give higher priority to Uncontrolled data traffic. The source node is allowed to set the priorities but on the way, routers can change it. Therefore, the destination should not expect the same priority which was set by the source node.

Flow Label (20-bits): Flow Label field is used by a source to label the packets belonging to the same flow in order to request special handling by intermediate IPv6 routers, such as non-default quality of service or real-time service. In order to distinguish the flow, an intermediate router can use the source address, a destination address, and flow label of the packets. Between a source and destination, multiple flows may exist because many processes might be running at the same time. Routers or Host that does not support the functionality of flow label field and for default router handling, flow label field is set to 0. While setting up the flow label, the source is also supposed to specify the lifetime of the flow.

Payload Length (16-bits): It is a 16-bit (unsigned integer) field, indicates the total size of the payload which tells routers about the amount of information a particular packet contains in its payload. The payload Length field includes extension headers(if any) and an upper-layer packet. In case the length of the payload is greater than 65,535 bytes (payload up to 65,535 bytes can be indicated with 16-bits), then the payload length field will be set to 0 and the jumbo payload option is used in the Hop-by-Hop options extension header.

Next Header (8-bits): Next Header indicates the type of extension header(if present) immediately following the IPv6 header. Whereas In some cases it indicates the protocols contained within upper-layer packets, such as TCP, UDP.

Hop Limit (8-bits): Hop Limit field is the same as TTL in IPv4 packets. It indicates the maximum number of intermediate nodes IPv6 packet is allowed to travel. Its value gets decremented by one, by each node that forwards the packet and the packet is discarded if the value decrements to 0. This is used to discard the packets that are stuck in an infinite loop because of some routing error.

Source Address (128-bits): Source Address is the 128-bit IPv6 address of the original source of the packet.

Destination Address (128-bits): The destination Address field indicates the IPv6 address of the final destination(in most cases). All the intermediate nodes can use this information in order to correctly route the packet.

Extension Headers: In order to rectify the limitations of the IPv4 Option Field, Extension Headers are introduced in IP version 6. The extension header mechanism is a very important part of the IPv6 architecture. The next Header field of IPv6 fixed header points to the first Extension Header and this first extension header points to the second extension header and so on.

- **ICMP**

Internet Control Message Protocol (ICMP) is a network layer protocol used to diagnose communication errors by performing an error control mechanism. Since IP does not have an inbuilt mechanism for sending error and control messages. It depends on Internet Control Message Protocol(ICMP) to provide [error control](#).

ICMP is used for reporting errors and management queries. It is a supporting protocol and is used by network devices like routers for sending error messages and operations information. For example, the requested service is not available or a host or router could not be reached.

Uses of ICMP

ICMP is used for error reporting if two devices connect over the internet and some error occurs, So, the router sends an ICMP error message to the source informing about the error. For Example, whenever a device sends any message which is large enough for the receiver, in that case, the receiver will drop the message and reply back ICMP message to the source.

Another important use of ICMP protocol is used to perform network diagnosis by making use of traceroute and ping utility. We will discuss them one by one.

Traceroute: [Traceroute](#) utility is used to know the route between two devices connected over the internet. It routes the journey from one router to another, and a traceroute is performed to check network issues before data transfer.

Ping: [Ping](#) is a simple kind of traceroute known as the echo-request message, it is used to measure the time taken by data to reach the destination and return to the source, these replies are known as echo-replies messages.

- **IGMP**

IGMP is acronym for Internet Group Management Protocol. IGMP is a communication protocol used by hosts and adjacent routers for multicasting communication with IP networks and uses the resources efficiently to transmit the message/data packets. Multicast communication can have single or multiple senders and receivers and thus, IGMP can be used in streaming videos, gaming or web conferencing tools. This protocol is used on IPv4 networks and for using this on IPv6, multicasting is managed by Multicast Listener Discovery (MLD). Like other network protocols, IGMP is used on network layer. MLDv1 is almost same in functioning as IGMPv2 and MLDv2 is almost similar to IGMPv3. The communication protocol, IGMPv1 was developed in 1989 at Stanford University. IGMPv1 was updated to IGMPv2 in year 1997 and again updated to IGMPv3 in year 2002.

Applications:

Streaming – Multicast routing protocol are used for audio and video streaming over the network i.e., either one-to-many or many-to-many.

Gaming – Internet group management protocol is often used in simulation games which has multiple users over the network such as online games.

Web Conferencing tools – Video conferencing is a new method to meet people from your own convenience and IGMP connects to the users for conferencing and transfers the message/data packets efficiently.

Shortest path algorithm:

In this algorithm, to select a route, the algorithm discovers the shortest path between two nodes. It can use multiple hops, the geographical area in kilometres or labelling of arcs for measuring path length.

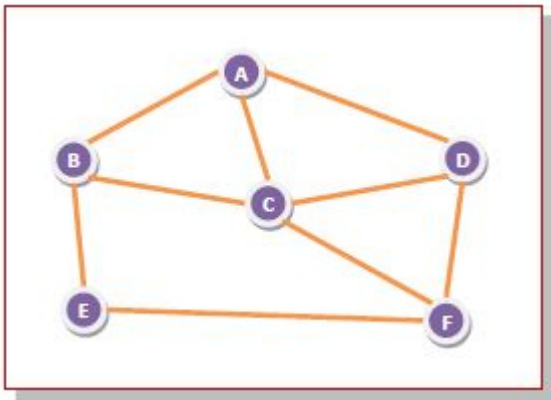
The labelling of arcs can be done with mean queuing, transmission delay for a standard test packet on an hourly basis, or computed as a function of bandwidth, average distance traffic, communication cost, mean queue length, measured delay or some other factors.

In shortest path routing, the topology communication network is defined using a directed weighted graph. The nodes in the graph define switching components and the directed arcs in the graph define communication connection between switching components. Each arc has a weight that defines the cost of sharing a packet between two nodes in a specific direction.

This cost is usually a positive value that can denote such factors as delay, throughput, error rate, financial costs, etc. A path between two nodes can go through various intermediary nodes and arcs. The goal of shortest path routing is to find a path between two nodes that has the lowest total cost, where the total cost of a path is the sum of arc costs in that path.

Flooding is a non-adaptive routing technique following this simple method: when a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.

For example, let us consider the network in the figure, having six routers that are connected through transmission lines.



Using flooding technique –

- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C and E.

Types of Flooding

Flooding may be of three types –

- **Uncontrolled flooding** – Here, each router unconditionally transmits the incoming data packets to all its neighbours.
- **Controlled flooding** – They use some methods to control the transmission of packets to the neighbouring nodes. The two popular algorithms for controlled flooding are Sequence Number Controlled Flooding (SNCF) and Reverse Path Forwarding (RPF).
- **Selective flooding** – Here, the routers don't transmit the incoming packets only along those paths which are heading towards approximately in the right direction, instead of every available paths.

